APPENDIX G: ALGORITHM SELECTION AND COMPARISON

G.1 Algorithm Used in the Project

The implemented predictor uses sliding-window ordinary least-squares linear regression. Separate straight-line models are fitted to recent latitude, longitude and altitude observations as functions of time. The method is an interpretable short-horizon baseline; it does not claim to reconstruct complete flight plans.

Implemented constraints

- Up to 20 recent samples are read; at least 4 valid time-ordered samples are required.
- Predictions are produced only after an aircraft disappears from the current session snapshot.
- The maximum prediction age is 180 seconds; older records are omitted.
- Constant speed, heading and vertical rate provide a deterministic fallback.
- Predicted rows are output-only and are never inserted into the real trajectory table.

G.2 Selection Rationale

Linear regression is computationally light, explainable and independent of a large training dataset. Its limitation is the local straight-motion assumption. Kalman filtering offers stronger state/noise modelling [10], while random forests [11] and LSTM-based models [12]-[14] require representative training data and more extensive validation. The current method is therefore appropriate as a transparent baseline for short receiver gaps.



Figure 14: Trajectory Prediction Pipeline